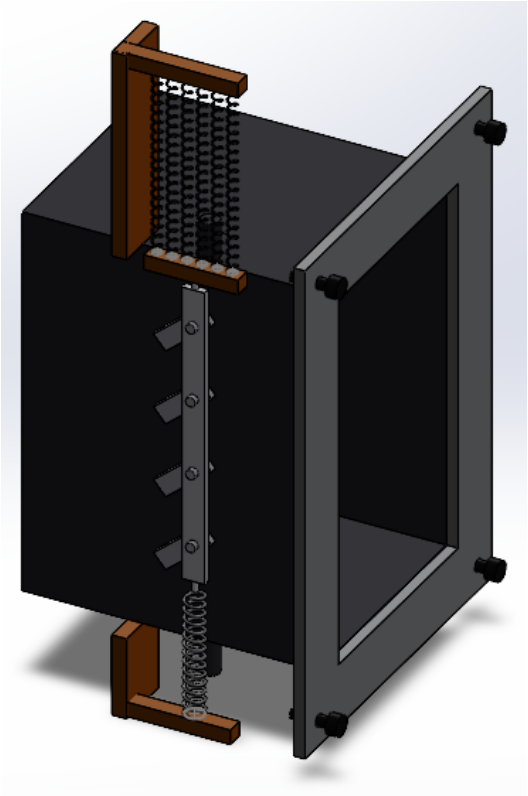
Project Guidebook: Passive SMA Brake-Duct Actuator

What it is: a brake-cooling duct whose louvers open and close by themselves, driven by shape-memory-alloy springs heated by the brake disc. There are no sensors, wires, motors or power supply in the device.

What this guidebook covers: the problem, how the device works, how each part of the project was designed and checked, which tools were used and why, what was found (including what did not work), and how to reproduce everything.



The actuator module

1. The problem

A Formula 1 brake disc reaches about 800-1000 °C under heavy braking. The brakes are cooled by ducts that scoop air from in front of the wheel. Ducts have a fixed size, so they are a compromise:

- **Under braking** the disc needs as much air as possible, or the friction material overheats and wears.
- **On the straights** the brakes are idle, so cooling air is wasted. The air pushed through the duct costs aerodynamic drag.

The idea: **close the duct on the straights and open it when the brakes are hot**, with no electronics. The design brief set three targets:

Target in the brief	Value
Drag saved on straights by closing the duct	-18.4 %
Temperature at which the louvers start to open	85 °C
Rod stroke for full opening	14.2 mm

Part of this project's value is that it **checks the brief instead of assuming it**. Two of the three targets held up; the drag target did not (section 5.8).

2. How the device works

2.1 Shape-memory alloy in one paragraph

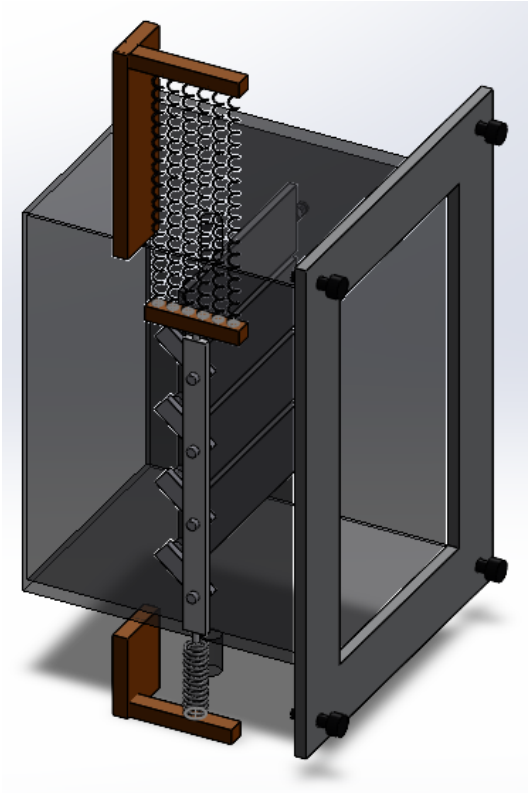
Nickel-titanium (Nitinol) has two crystal structures. Below a transformation temperature it is soft **martensite** and can be stretched easily. Heated past the **austenite finish temperature (Af)**, it returns to its remembered shape and pulls with a large force. On cooling past the **martensite finish temperature (Mf)** it becomes soft again. The gap between opening and closing temperatures is the **hysteresis**. A one-way SMA spring cannot push itself back, so a steel **bias spring** stretches it again when it cools.

2.2 The mechanism

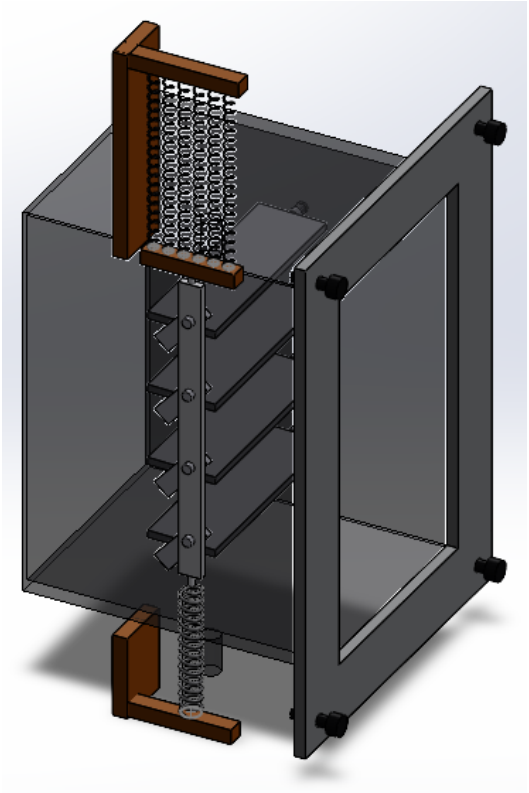
- Four louvers sit across the duct throat, each on a pivot shaft.
- Each louver has a small external **crank arm** (radius 10.89 mm). A single **linkage rod** joins the four crank pins, so all louvers move together.
- A pack of **six NiTiCu springs** pulls the rod up when hot. The **bias spring** pulls it down when cold.
- Rod stroke and louver angle are linked by the crank geometry:

`stroke = R (sin(θ + ϕ0) - sin ϕ0)`, with R = 10.89 mm and φ0 = -40°

This gives 14.0 mm of stroke at 80° of louver rotation, and 15.15 mm at 88.5°.



Cold: louvers closed



Hot: louvers open 88.5°

2.3 What heats the spring

The project found that **hot air alone cannot do it**; the air near the duct stays far too cool (section 5.2). The spring is heated by **thermal radiation from the glowing disc**. It is coated black (emissivity about 0.8) to absorb that radiation, and is cooled by the airflow when the car is fast.

3. Results at a glance

Question	Answer
Does the linkage give the stroke?	Yes: 14.000 mm at 80° (SOLIDWORKS kinematics)
Can hot air trigger the spring?	No: 300-324 K against 358 K needed (Fluent CHT)
Can rotor radiation trigger it?	Yes: 119-129 kW/m² on the spring; a dark wire reaches 378-414 K
Does it close again on the straight?	Yes: wire settles at 306-311 K, below Mf
Does the brief's 1.2 mm NiTi spring work over a lap?	No: it never opens, or stays stuck open
Fix	Six 0.5 mm NiTiCu springs (Af 76 °C / Mf 58 °C): open after the 3 hardest braking zones, open only 2 % of straight time
Is it strong enough?	Yes: all stress checks pass, lowest factor of safety 3.3 (MAPDL)
Does closing the duct save 18.4 % drag?	**No. The duct's own momentum drag is only 1.1-1.4 % of the total, and with a rotating wheel and moving road closing actually adds 0.5 % drag**
Is it legal in F1?	No: moving aerodynamic parts are banned
Can it be tested?	A bench rig, its electronics, firmware and a digital twin are designed; the twin predicts the rig reproduces the car behaviour
Where does the idea pay off?	An EV grille shutter: closing it cuts drag 14.5 % on a simple CFD car; on a real car about 3-5 % less battery energy, with no motor or wiring

4. Tools used and why

Tool	Used for
SOLIDWORKS 2025	All CAD: parts, assemblies, mates, kinematic checks, interference checks, CFD domains, drawings, renders
ANSYS Fluent 2026 R1 (Student)	Conjugate heat transfer with radiation; isolated-wheel aerodynamics
ANSYS MAPDL 2026 R1 (Student)	Linkage mechanism and spring force model; stress checks with contact
gmsh	Tetrahedral meshing of parts for MAPDL
Python	Heat-balance and transient lap models, data logger, post-processing, plots, reports
KiCad 10	Schematic, PCB layout, design-rule checks, Gerbers, 3D board model
Arduino	Firmware for the test rig's controller (Arduino Uno)
C++	Firmware logic, also compiled for the PC to run the digital twin

All models, settings and results are kept in the repository, so every result can be reproduced.

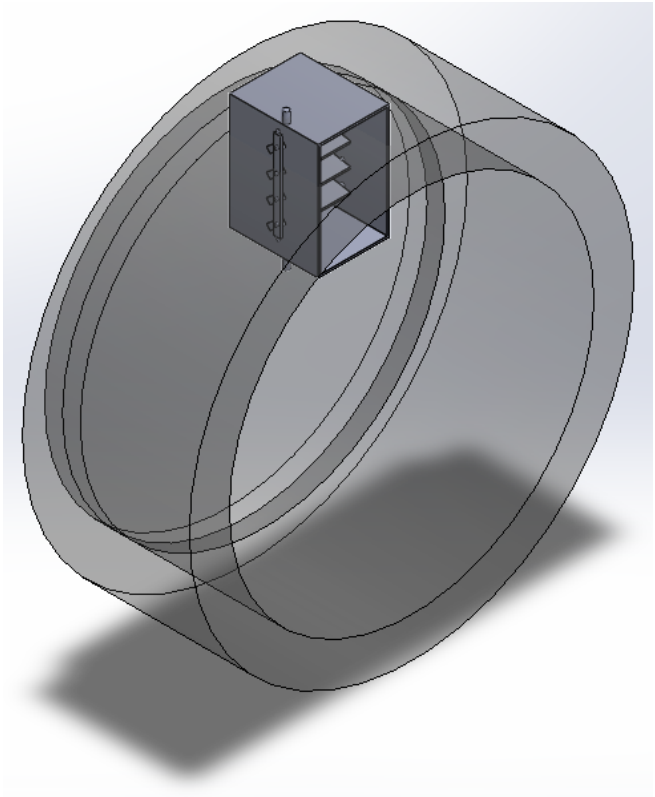
5. The work, phase by phase

5.1 Phase 1 - CAD and kinematics (SOLIDWORKS)

Built: the packaging zone around a 328 mm disc and a 400 mm rim. A 100 x 60 mm duct throat (80 mm long, 2 mm carbon wall). Four 56 x 18 x 2 mm louvers on 20 mm pitch with external cranks, and the linkage rod. 16 mates.

Checked: a temporary angle mate drove louver 1 through 0-80°, and the rod position was read at each step. The measured stroke matched the crank formula to 0.001 mm (14.000 mm at 80°).

Bug found and fixed later: concentric mates only align axes, so the rod had solved on the wrong side of the duct, about 70 mm from the crank pins. The assembly now checks the rod position and flips the distance mate if needed.



Phase 1 assembly

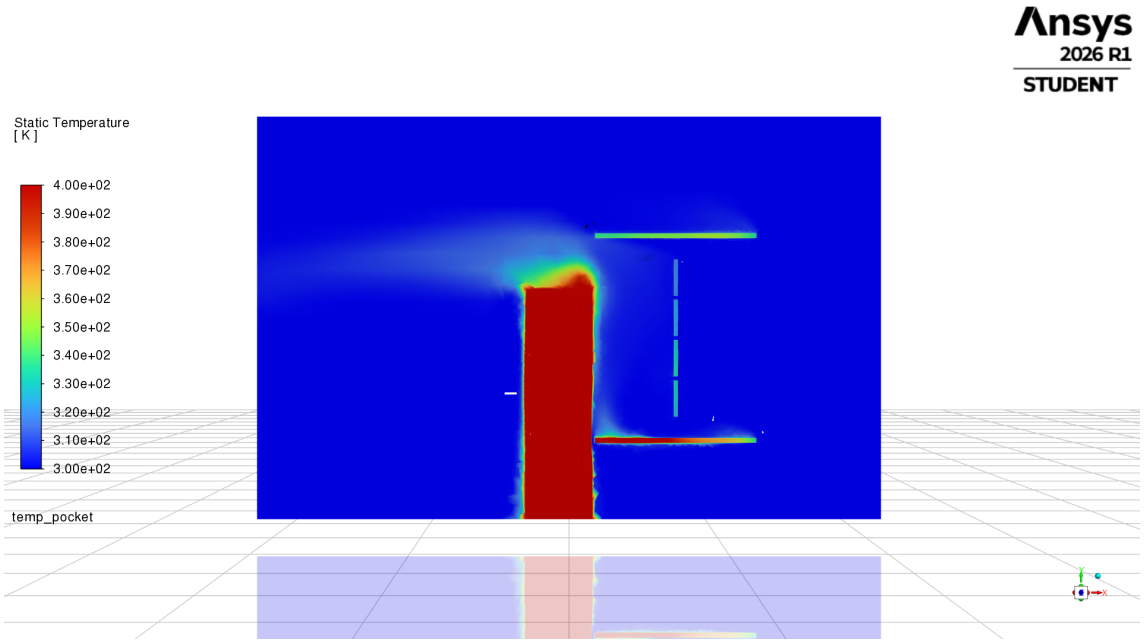
5.2 Phase 2 - Can the spring be triggered? (Fluent CHT)

Model: a box around a rotor segment, the duct, the louvers and a pocket of air where the spring sits. Solids and air are solved together (conjugate heat transfer), with k-ω SST turbulence and Discrete Ordinates radiation. Meshes are about 0.5 million cells (the Student licence caps at 1 million).

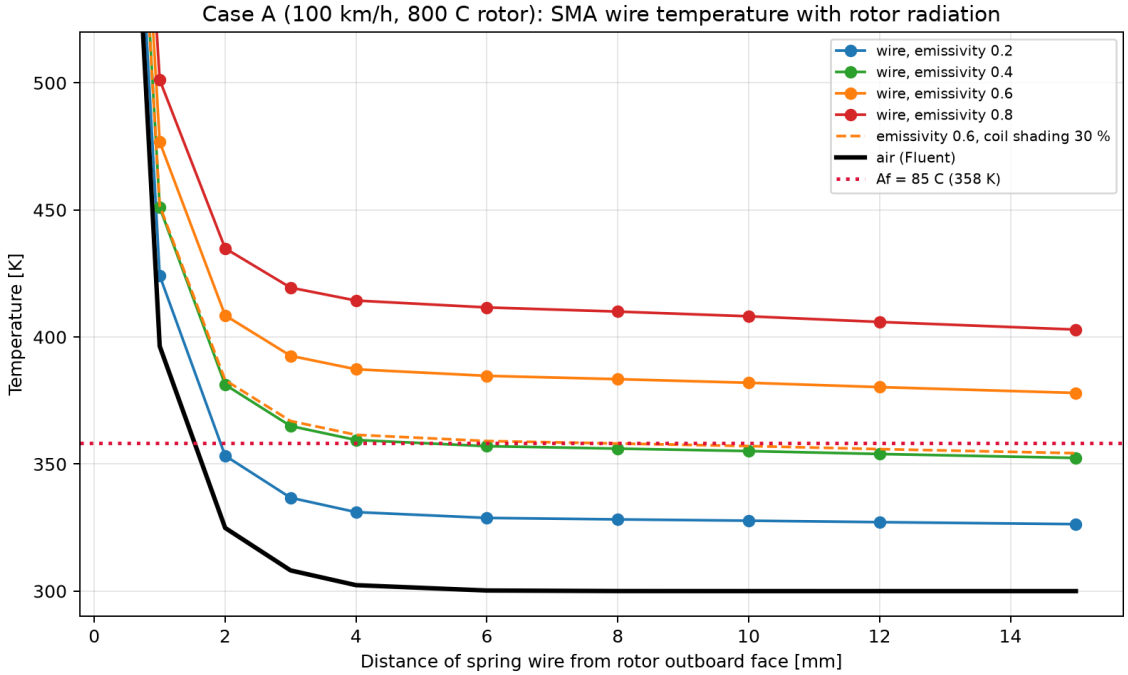
- **Case A, braking:** 100 km/h, 800 °C rotor, louvers closed.

- Spring pocket beside the duct: 300.3 K.
- Pocket moved to the rotor's wake face: 324.1 K.
- Both are well short of the original spring's 358 K (85 °C). The hot air is only a ~2 mm layer against the disc.

- **Radiation check:** the disc puts 119-129 kW/m² of radiation on the relocated pocket. A heat balance on the wire, with absorbed radiation against re-radiation and convection, gives 378-387 K for a wire emissivity of 0.6 and 403-414 K at 0.8.
  - **Conclusion:** the spring works if it is dark (emissivity of at least about 0.45).
- **Case B, straight:** 300 km/h, 400 °C rotor, louvers open. Radiation falls to about 20 kW/m² and airflow doubles. The wire settles at 306-311 K, below Mf, so **the louvers close again**.



Case A pocket temperature



Wire temperature vs emissivity

5.3 Phase 2 - Does it work over a whole lap? (transient model)

Steady cases are not enough: brake temperatures change every few seconds. A reduced-order lap model:

- **Lap trace:** a synthetic 98.5 s lap with five braking zones, 300 to 100 km/h.

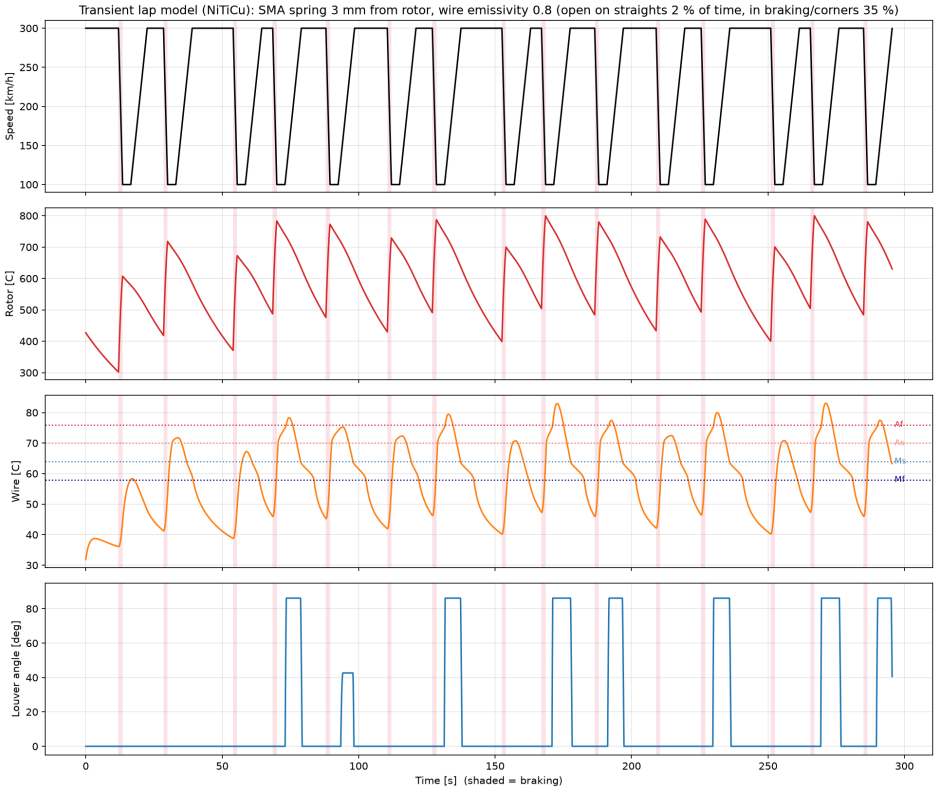
- **Rotor:** heat balance fitted to the brief (800 °C peak, 400 °C on the longest straight).
- **Spring environment:** radiation and airflow at the spring interpolated between the two Fluent cases.
- **Wire physics:** wire heat balance with latent heat, Liang-Rogers transformation kinetics, and stroke from the MAPDL sweep.

Spring	Result over a lap
Brief: NiTi 1.2 mm, Af 85 °C	Never opens (emissivity 0.6), or opens then stays open on 91 % of straights (0.8)
Redesign: NiTiCu 0.5 mm, Af 76 / As 70 / Ms 64 / Mf 58 °C, emissivity 0.8	**Opens 4-5 s after the 3 hardest braking zones; open only 2 % of straight time**

Why the redesign works:

- **Faster response:** a 1.2 mm wire takes about 5 s to respond, while the rotor peak lasts only a moment. A 0.5 mm wire responds in about 1.4 s.
- **Narrower hysteresis:** NiTi's ~40 K hysteresis is wider than the wire's temperature swing over a lap, so it can never cycle. NiTiCu's ~12 K fits.

The cost is force: one thin spring gives about 5.6 N, so six are used in parallel.

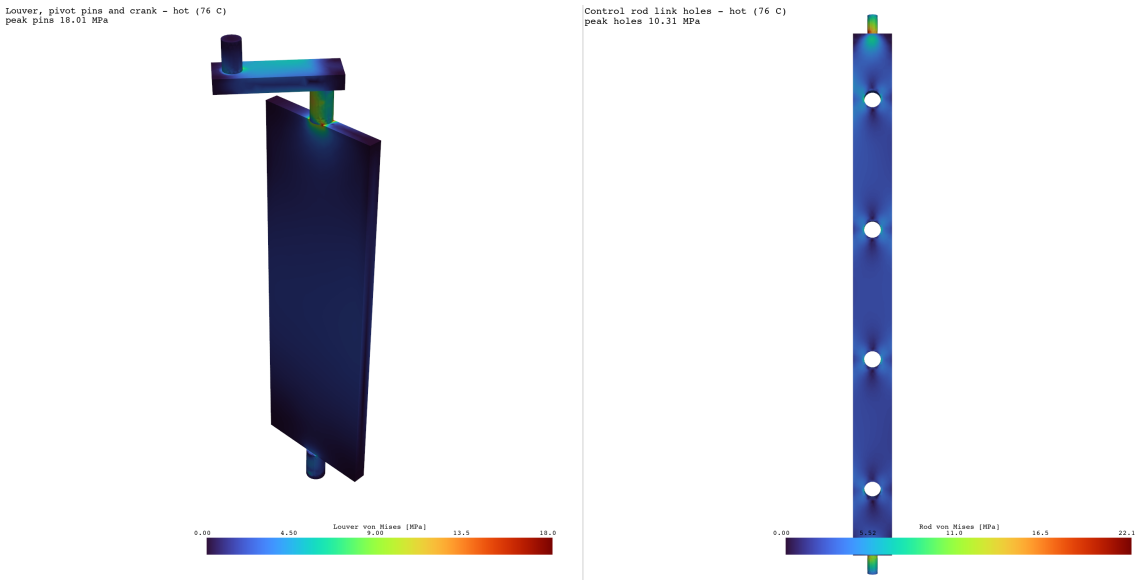


NiTiCu spring over three laps

5.4 Phase 3 - Structure and force (MAPDL)

- **Mechanism model:** beam elements for the linkage, plus a nonlinear spring element carrying the SMA's temperature-dependent force against the bias spring and the aerodynamic torque on the louvers.
- **Stress model:** solid tetrahedral meshes of the louver, pins, rod and housing holes, with contact. Each joint is checked against a factor of safety of 2.5.

	Brief spring	Six-spring NiTiCu pack
Hot state	86.2° / 14.85 mm at 85 °C	88.5° / 15.15 mm at 76 °C
Louver pins, hot	17.2 MPa, FoS 8.1	18.0 MPa, FoS 7.8
Lowest FoS anywhere	3.3	3.3



Stress, hot state

5.5 Phase 4 - Data logger

A Python logger reads one JSON line per sample at 20 Hz:

- **Recording:** it logs temperature, force and stroke, and draws a live dashboard.
- **Correlation:** it compares each sample with the FEA prediction and reports onset temperature, hysteresis width, stroke and force error.
- **Connection:** it works with a real serial port or, unchanged, with the digital twin over a TCP "serial port" (section 5.7).

5.6 Phase 5 - Technical whitepaper

The whitepaper is a 6-page PDF assembled directly from the result files.

- **Labelling:** every value is marked as measured, simulated or design target.
- **Content:** it includes the FIA regulation assessment.

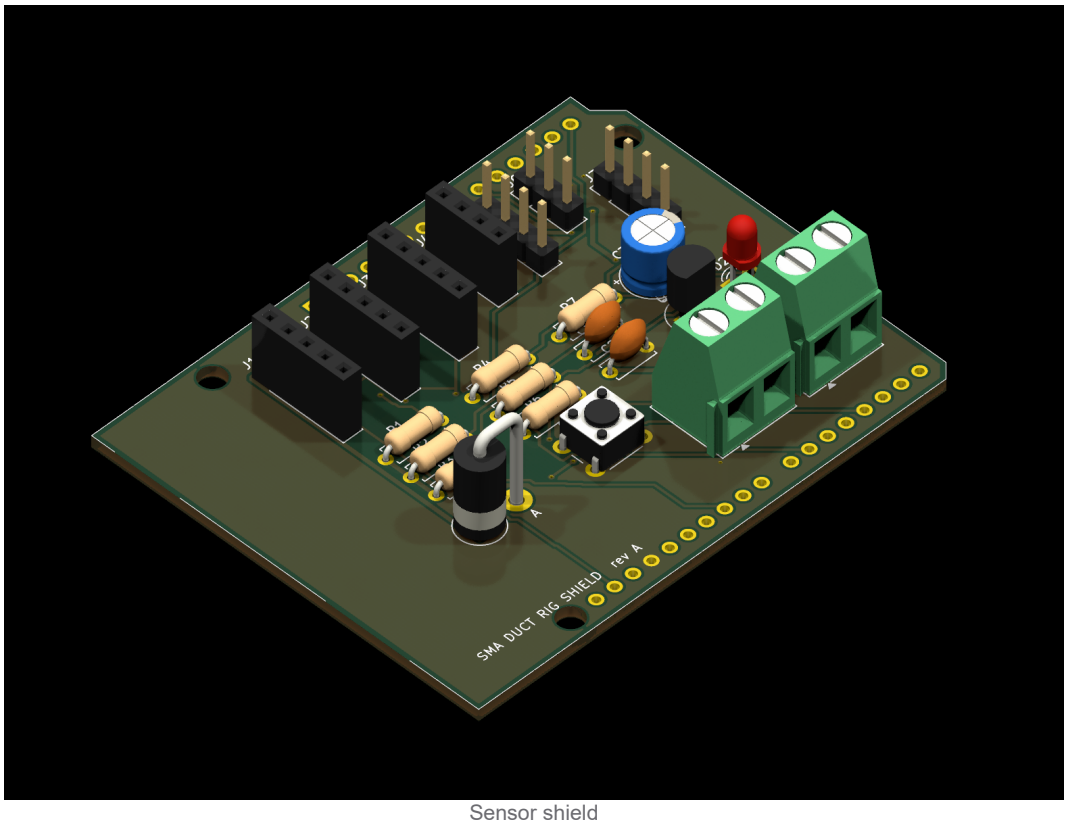
5.7 Phase 6 - Bench rig, electronics, firmware and digital twin

The rig (SOLIDWORKS, zero interferences):

- **Rotor stand-in:** a 500 W halogen lamp in a reflector, 23 mm from the springs (about 143 kW/m²).
- **Car-speed stand-in:** a 50 mm blower aimed at the springs (about 13 m/s).
- **Duct flow:** a 92 mm fan feeding the duct through a printed plenum.
- **Sensors:** thermocouples, a magnetic angle sensor on a louver shaft, and a load cell on the spring pack.

The electronics (KiCad): an Arduino Uno shield.

- **Parts:** three thermocouple amplifiers, the angle sensor, a load-cell amplifier, a pitot sensor, 25 kHz fan PWM and a solid-state-relay driver for the lamp.
- **Checks:** schematic rule check 0 violations; PCB design-rule check 0 errors and 0 unconnected nets; Gerbers ready to order.



Sensor shield

The firmware (Arduino Uno: 57 % flash, 55 % RAM):

- **Telemetry:** 20 Hz output in the logger's format.
- **Commands:** heat, fan, PI temperature hold and ramp, and replay of the car lap.
- **Safety:** cut-outs for spring over-temperature, reflector over-temperature and a broken thermocouple.

The digital twin: the same firmware logic is compiled as a PC library and driven by a physics model of the rig:

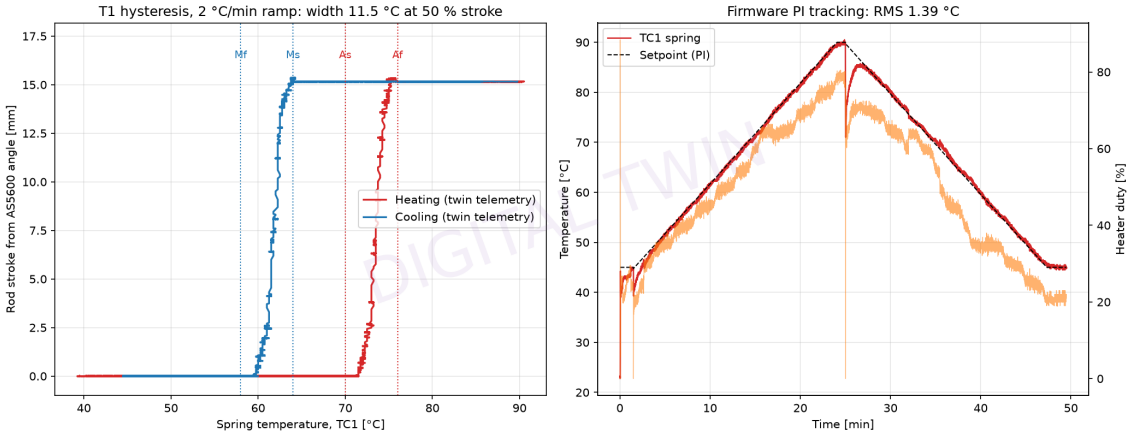
- lamp, reflector, wire heat balance, SMA kinetics and the MAPDL stroke curve;
- emulated sensors with realistic update rate, lag, resolution and noise.

The unchanged logger connected to it and logged 2,400 samples at 20.0 Hz.

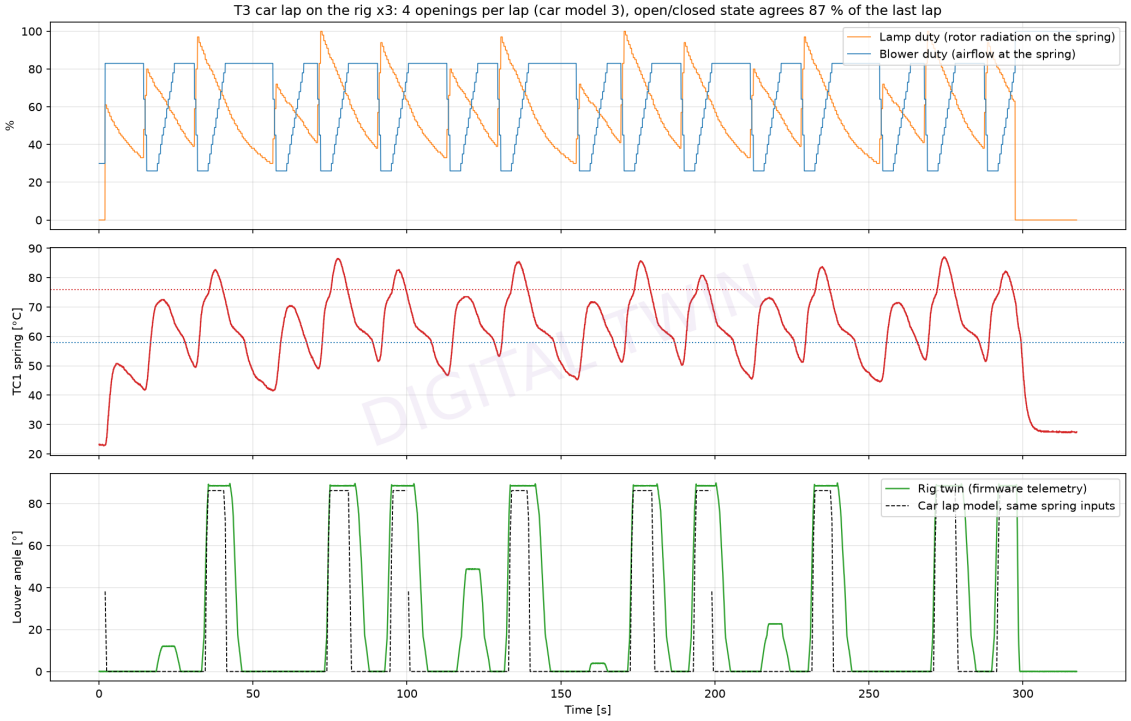
Virtual test	Result
Slow ramp, 2 °C/min	Opens 72-75 °C, closes 63-60 °C, hysteresis 11.5 °C (design 12 °C)
Step response	0.5 mm wire opens in 3.6 s, closes in 1.9 s; 1.2 mm wire closes in 5.7 s
Car lap replayed on the rig	Same 3 openings per lap as the car model; open/closed state agrees 87 % of the lap
1,000 heat/cool cycles	Every cycle fully open and closed, no firmware faults
Faults	Broken thermocouple or failed blower: lamp latched off

Design problems caught before building:

1. **Heater window too slow.** A 1 s lamp on/off window made the thin wire ripple several degrees, and the measured hysteresis shrank to 4.6 °C. It is now 200 ms.
2. **First rig too weak.** It could not reproduce the car's peak radiation or airflow. The lamp moved closer, the blower was added, and the 12 V diode was upgraded from 1 A to 3 A.
3. **Quick tests misread hysteresis.** A quick 60 s heat / 60 s cool test reads 3.1 °C, because the thermocouple lags. Transformation temperatures must come from the slow ramp test.
4. **Ratcheting hysteresis model.** The textbook partial-loop rule drifted under small temperature ripple. It was replaced with a return-point-memory model. Re-running the lap model with it gave identical results.



Twin: hysteresis test



Twin: car lap replayed on the rig

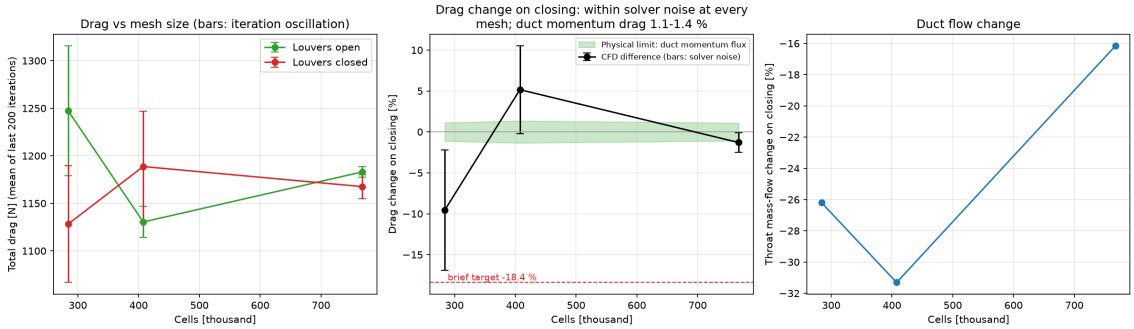
5.8 The drag question (Fluent, isolated wheel)

First attempt: a duct-sized box gave -2.0 %. It was rejected because the rotor blocked about 55 % of the box, so the box behaved like a narrow wind tunnel.

Second attempt: a 2.0 x 1.36 x 3.7 m domain around a 720 mm wheel with cover, rotor and the duct with a forward-facing scoop. 300 km/h, k-ω SST.

Mesh-independence study: the case was solved on three meshes, with all mesh sizes scaled together and the same boundary-layer cells. The Student licence's 1 million cell limit rules out a finer level.

Mesh	Cells	Drag open	Drag closed	Change on closing	Solver noise	Momentum limit
Coarse	284k	1,247 N	1,128 N	-9.6 %	±7.4 %	1.1 %
Medium	408k	1,130 N	1,189 N	+5.2 %	±5.4 %	1.4 %
Fine	768k	1,183 N	1,168 N	-1.3 %	±1.2 %	1.1 %



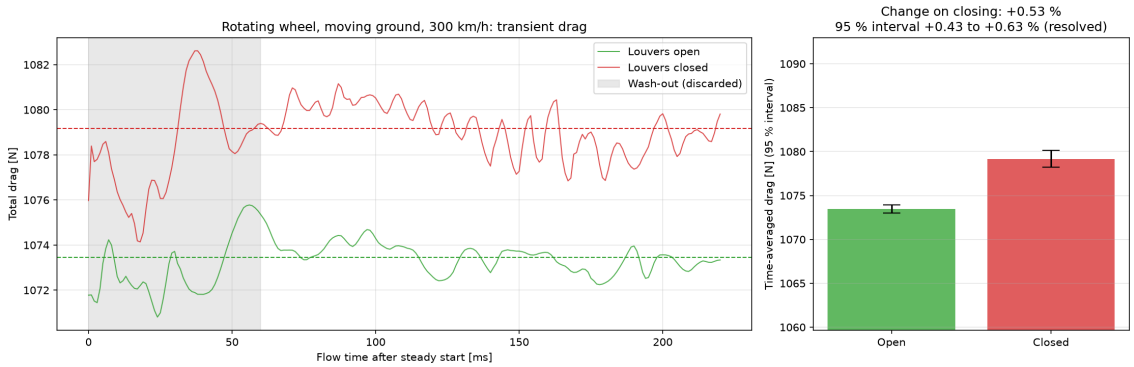
Mesh-independence study

What the study shows:

- **The direct CFD difference is not resolved.** It changes sign between meshes and stays within the solver's iteration noise every time.
  - The duct's own drag changes by only 1-2 N between open and closed.
  - The total swings come from the wheel's unsteady wake, which a steady simulation cannot settle.
- **The scale of the effect.** The duct's own through-flow (momentum) drag is the momentum of the air it swallows:
  `D_duct ≈ ṁ V = 0.155-0.183 kg/s x 83 m/s ≈ 13-15 N`
  That is **1.1-1.4 % of the total drag on all three meshes**. It is not a hard cap on the change: closing an inlet also changes how the outside flow meets the body (interference), and that part can add to or cancel the momentum term. The EV shutter study in this project measured interference at about four times the momentum term. The brief's -18.4 % would need interference more than ten times the momentum term, which is why the transient run below was needed to settle it.
- **Resolving the actual difference** needs time-averaged transient CFD with a rotating wheel on a moving ground (next step).

Third step: rotating wheel, moving road, time-averaged. The wheel, cover and disc spin at 231.5 rad/s (300 km/h on a 0.36 m tyre) and the ground moves at 300 km/h, as a real road does under the car. After a steady start, the flow is solved in 1 ms time steps and the drag is averaged over the settled 160 ms (fine mesh).

	Louvers open	Louvers closed
Steady drag	1,071.8 N	1,076.0 N
Time-averaged drag (± standard error)	1,073.5 ± 0.2 N	1,079.2 ± 0.5 N
Change on closing		**+0.53 % (95 % interval +0.43 to +0.63 %)**



Rotating wheel, transient drag

- **Closing the louvers does not save drag in this model; it adds about 0.5 %.**
  - The duct's own drag falls by only 0.4 N, while the wheel's drag rises by more.
  - The two drag traces stay apart for the whole averaging window, and the steady runs agree on the direction (+0.4 %).

- **Caveats:** the interval covers sampling noise only. 160 ms is about four wake cycles on one mesh, and the drag coefficient (1.19) is still above real-wheel values.

Conclusion of the drag study: in every model the effect of closing the louvers is a fraction of a percent, the same order as the duct's 1.1-1.4 % momentum drag. In the most realistic model it is a small penalty, not a saving. The brief's -18.4 % is ruled out.

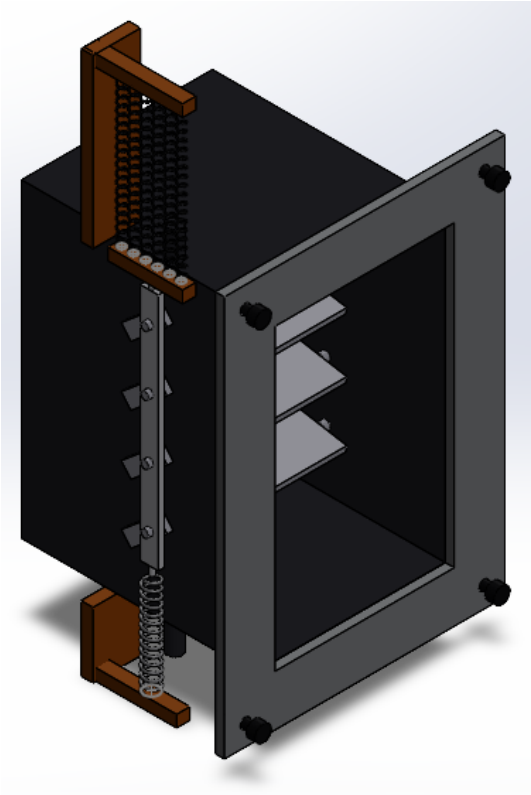
5.9 Phase 7 - Product package

Module CAD: 21 components, zero interferences.

- real helical NiTiCu springs with a black coating, and a steel bias spring;
- a rod crosshead, anodised brackets, and an inlet flange with M3 screws;
- materials assigned to every part.

Supporting assets:

- **Animation:** the springs genuinely stretch and contract as the louvers open and close.
- **Drawing SMA-DM-001:** A3, with a parts list.
- **Product sheet:** one page.



Module, isometric view

5.10 Re-target - passive SMA grille shutter for an electric car

F1 bans moving aero parts, and the brake duct moves too little air to matter. An EV radiator moves about ten times more air relative to the car's drag, and its coolant runs at 25-40 °C. That suits a low-temperature NiTi spring (As 35 / Af 40 °C) sitting in the coolant return line. Full details are in EV_COOLING.md.

CFD (half car, porous radiator, moving road):

- Closing the shutters cuts drag by 105 N at 100 km/h and 182 N at 130 km/h, 14.5 % both times (ΔCd 0.10).
- Iteration noise is under ±10 N.
- The saving scales with V^2.
- The saving is about four times the radiator's momentum drag ($\dot{m}\,V$). The rest is interference from the plain floor exit, which a production car would be shaped to avoid.

Drive-cycle model (CFD flow coefficient, cooling drag in the road load):

Scenario	SMA shutter	Battery energy saved, production-car estimate (κ = 1)
City 25 °C	stays shut	3.4 %
Motorway 130 km/h, 30 °C	stays shut, coolant 37 °C	5.4 % (motorised thermostat: 4.5 %)
Hot day 38 °C	opens fully	0 %, same coolant peak as an open grille

Module CAD: six louvers on a glass-filled PA66 frame driven through a scotch-yoke bar (14.8 mm lift for 80°).

- Five NiTi springs in a clear coolant sleeve lift the bar.
- A steel bias spring pulls it shut.
- Zero interferences shut and fully open.



EV shutter closed and open

6. Key equations

Wire heat balance (per unit surface, used for the steady check, the lap model and the twin):

```
ρ c_eff (D/4) dT/dt = ε G / 4 - ε σ (T⁴ - T_amb⁴) - h (T - T_air)
```

- G = incident radiation
- ε = wire emissivity
- h = convection coefficient (Churchill-Bernstein)
- c_eff = specific heat including the latent heat of transformation

Liang-Rogers kinetics (martensite fraction ξ):

- Heating, As < T < Af: `ξ = (ξ0 / 2) [cos(π (T - As) / (Af - As)) + 1]`
- Cooling, Mf < T < Ms: `ξ = ((1 - ξ0) / 2) cos(π (T - Mf) / (Ms - Mf)) + (1 + ξ0) / 2`

Crank kinematics: `stroke = R (sin(θ + ϕ0) - sin ϕ0)`

Duct momentum drag: `D_duct ≈ ṁ V`. This is the through-flow part only. Interference with the outside flow adds to it or cancels it, so the total cooling drag is written `κ ṁ V`, with κ found by CFD.

7. Limits

- **Not built:** no physical hardware exists yet. Rig and twin results are predictions.
- **Student licence:** meshes are capped at 1 million cells, so the CFD is coarse.
  - Thermal-case continuity stalled near 1e-2.
  - The rotating-wheel drag result is averaged over only 160 ms on one mesh.
- **Lap model:** reduced-order, on a synthetic lap; the NiTiCu transformation temperatures are design targets, not a datasheet alloy.
- **Rig assumptions:** the twin's lamp irradiance, blower curve and heat-transfer coefficients must be calibrated on a real rig.
- **Regulations:** F1 rules ban movable aerodynamic parts, so this is a research and non-FIA concept.

8. Next steps

1. Order the shield PCB and rig parts, and source NiTiCu springs to the 76 / 58 °C specification.
2. Build the rig, calibrate it, and repeat the five virtual tests on real hardware.
3. Compare measured hysteresis, response time and force with the predictions, and update the models.
4. Re-target the concept where it is legal and useful (started: the EV grille shutter in section 5.10). Next for it: a production-shaped cooling exit in the CFD, a coolant-sleeve bench test, and sourcing a 35 / 40 °C NiTi grade.

9. Repository map

Folder	Contents
`phase1_*.py`, `swlib.py`	Duct, louvers, rod and assembly in SOLIDWORKS
`phase2_*.py`	CFD domains, meshing, solves, radiation, wire balance, lap model, wheel aero
`phase3_fea.py`	MAPDL mechanism and stress model
`phase4_bench_validation.py`	Data logger and correlation
`phase5_generate_whitepaper.py`	Whitepaper PDF
`rig_cad.py`, `electronics/`, `firmware/`	Bench rig CAD, shield PCB, firmware, digital twin
`product_*.py`	Product CAD, animation, drawing, product sheet
`ev/`	EV grille-shutter re-target: sizing, half-car CFD, drive-cycle model, shutter CAD
`cad/`, `cfd/`, `fea/`, `bench/`, `report/`	Models, plots, result files, PDFs
`docs/`	This guidebook, detailed results, rig and product notes

10. Glossary

Term	Meaning
Af / As	Austenite finish / start: temperatures at which the spring finishes / starts recovering its shape
Mf / Ms	Martensite finish / start: temperatures at which it finishes / starts becoming soft again
Hysteresis	Gap between opening and closing temperatures
Bias spring	Steel spring that re-stretches the cold SMA spring
CHT	Conjugate heat transfer: heat in solids and fluid solved together
DO radiation	Discrete Ordinates: Fluent's radiation model
k-ω SST	Turbulence model suited to separated flow near walls
FoS	Factor of safety: material strength divided by working stress
Digital twin	Software copy of the rig that runs the real firmware against a physics model

Term	Meaning
ERC / DRC	Electrical / design rule check in KiCad
Gerbers	Manufacturing files a PCB fab uses to make the board
Emissivity	How well a surface absorbs and emits thermal radiation (0 to 1)